

PAST EXAMINATION QUESTIONS (FP2)

TOPIC 9: HYPERBOLIC FUNCTIONS

1. (a) Sketch graph of $y = \coth x$ for $x > 0$ and state the equations of the asymptotes. [2]
 (b) Starting from the definition of $\coth$ and cosech in terms of exponentials, prove that

$$\coth^2 x - \operatorname{cosech}^2 x = 1. \quad [3]$$

Oct/Nov 2020/21/23/8 (part)

2. (a) Starting from the definitions of $\tanh$ and sech in terms of exponentials, prove that

$$1 - \tanh^2 \theta = \operatorname{sech}^2 \theta. \quad [3]$$

May/June 2020/23/6 (part)

3. (a) Starting from the definition of $\tanh$ in terms of exponentials, prove that $\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$. [3]

Specimen Paper 2020/2/7 (part)

4. (i) Use the definition $\tanh y = \frac{e^{2y}-1}{e^{2y}+1}$ to show that $\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$ for $|x| < 1$. [3]
 (ii) Solve the equation $\tanh x + \coth x = 4$, giving your answer in the form $p \ln m$, where p is a positive rational number and m is a positive integer. $\left(\frac{1}{4} \ln 3 \right)$ [5]

CAIE 9795/May/June 2017/1/5

5. (i) (a) Show that $\tanh x = \frac{e^{2x}-1}{e^{2x}+1}$. [2]
 (b) Hence, or otherwise, show that, if $\tanh x = \frac{1}{k}$ for $k > 1$, then $x = \frac{1}{2} \ln \left(\frac{k+1}{k-1} \right)$ and find an expression in terms of k for $\sinh 2x$. $\left(\frac{2k}{k^2-1} \right)$ [4]

CAIE 9795/May/June 2014/1/12 (part)

6. (i) Use the definitions $\tanh \theta = \frac{e^\theta - e^{-\theta}}{e^\theta + e^{-\theta}}$ and $\operatorname{sech} \theta = \frac{2}{e^\theta + e^{-\theta}}$ to prove the results
 (a) $\tanh^2 \theta = 1 - \operatorname{sech}^2 \theta$,

CAIE 1348/May/June 2013/1/13 (part)

7. (i) On a single diagram, sketch the graphs of $y = \tanh x$ and $y = \cosh x - 1$, and use your diagram to explain why the equation $f(x) = 0$ has exactly two roots, where

$$f(x) = 1 + \tanh x - \cosh x. \quad [3]$$

CAIE 9795/May/June 2011/1/4 (part)

8. (i) Using the definitions of $\sinh$ and $\cosh$ in terms of exponentials, prove that

$$\cosh A \cosh B + \sinh A \sinh B \equiv \cosh(A + B). \quad [1]$$

- (ii) Solve the equation $5 \cosh x + 3 \sinh x = 12$, giving your answers in the form $\ln(p \pm q\sqrt{2})$ for rational numbers p and q to be determined. $(\ln(1.5 \pm \sqrt{2}))$ [5]

CAIE 9795/May/June 2010/1/4

9. Solve exactly the equation

$$5 \cosh x - \sinh x = 7,$$

giving your answers in logarithmic form. $\left(x = \ln 3, \ln \frac{1}{2}\right)$ [6]

CAIE 9795/Specimen Paper 2010/1/3

10. (b) Given that $4 \cosh x - \sinh x = 8$,
 (i) find the possible values of e^x , and hence find the possible values of $\tanh 2x$. [6]
 $\left(e^x = \frac{1}{3}, 5; \tanh 2x = -\frac{40}{41}, \frac{312}{313}\right)$

(ii) show that the only possible positive value of a for which $\tanh(ax) = \frac{9}{10}$ is $\frac{\ln 19}{\ln 25}$. [2]

Nov 1999/1/3 (part)

11. (a) (i) Prove that the principal value of $\cosh^{-1} x$, where $x \geq 1$, is $\ln[x + \sqrt{x^2 - 1}]$. [3]
 (ii) Hence, or otherwise, show that if α is positive and

$$\ln[2y + 3 + \sqrt{4y^2 + 12y + 8}] > \alpha,$$

then,

$$y > \frac{e^\alpha}{4} + \frac{1}{4e^\alpha} - \frac{3}{2}. \quad [3]$$

June 1993/1/3 (part)

12. Starting from the definition of $\sinh x$ and $\cosh x$ in terms of e^x and e^{-x} , prove the identity

$$\sinh(x - y) = \sinh x \cosh y - \sinh y \cosh x.$$

Show that

$$\tanh\left(n + \frac{1}{2}\right)x - \tanh\left(n - \frac{1}{2}\right)x = \sinh x \operatorname{sech}\left(n + \frac{1}{2}\right)x \operatorname{sech}\left(n - \frac{1}{2}\right)x. \quad [6]$$

December 1990/1/2 (part)

13. Starting from definitions of $\sinh x$ and $\cosh x$ in terms of exponentials, prove that

(i) $\sinh 2x = 2 \sinh x \cosh x$.

(ii) $\sinh^{-1} x = \ln[x + \sqrt{x^2 + 1}]$.

Solve the equation $\sinh 2x + \cosh x = 0$, giving your answer correct to three significant figures.

MB June 1985/1/20 (part)

14. (a) Find the exact real solutions of the simultaneous equations

$$\sinh x + \sinh y = \frac{25}{12},$$

$$\cosh x - \cosh y = \frac{5}{12}. \quad [5]$$

$(x = \ln 3, y = \ln 2)$
December 1984/1/17 (part)

15. (b) Find a set of values of the real constant a for which the equation

$$\sinh x + 2 \cosh x = a$$

is satisfied by two real distinct values of x . ($a > \sqrt{3}$)

Solve, correct to three significant figures, the equation

$$\sinh x + 2 \cosh x = 4.$$

(0.930, -2.03)
MB November 1983/1/20 (part)

16. (a) Given that $7 \sinh x + 20 \cosh x = 24$, find all the possible values of $\tanh \frac{1}{2}x$.
 $\left(-\frac{1}{2}, \frac{2}{11}\right)$

December 1982/1/16 (part)

17. Express $\tanh x$ in terms of e^{2x} and sketch the graph of $y = \tanh x$. $\left(\frac{e^{2x}-1}{e^{2x}+1}\right)$

Given that $\tanh(x+y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$, show that the $\tanh 3x = \frac{3 \tanh x + \tanh^3 x}{1 + 3 \tanh^2 x}$.

Solve the equation $\tanh 3x = 2 \tanh x$, giving your answers for x to two significant figures.
(0, ± 0.48)

MB November 1982/2/21

18. (a) Given that $\sinh x = \frac{1}{2}(e^x - e^{-x})$ and $\cosh x = \frac{1}{2}(e^x + e^{-x})$, prove the following results:

(i) $\cosh 2x = 2 \sinh^2 x + 1$,

(ii) $\sinh^{-1} x = \ln \left[x + \sqrt{x^2 + 1} \right]$.

Solve the equation $\cosh 2x = 4 + \sinh x$, giving the values of x correct to 2 significant figures. **(-0.88, 1.2)**

MB June 1981/1/20 (part)

19. (b) Starting with the definitions of $\sinh x$ and $\cosh x$ in terms of e^x and e^{-x} , show that $\sinh 2x = 2 \sinh x \cosh x$.

Hence, or otherwise solve the equation $\sinh 2x = 3 \cosh x$, expressing your answer in logarithmic form. **($\ln(1.5 + \sqrt{3.25})$)**

MB June 1980/2/20 (part)

20. (a) Solve the equation $2 \sinh x + 3 \cosh x = 4$, giving your answers to two decimal places. **(0.38, -1.99)**
 (b) Sketch the graphs of $y = \sinh^{-1} x$ and $y = \cosh^{-1} x$.

- (c) Prove that $\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$, $|x| < 1$. Sketch the graph of $y = \tanh^{-1} x$.

MB June 1979/1/20

21. (a) Given that $\cosh x = \frac{1}{2}(e^x + e^{-x})$ and $\sinh x = \frac{1}{2}(e^x - e^{-x})$, prove that $\cosh^2 x - \sinh^2 x = 1$, and derive an expression for $\cosh 2x$ in terms of $\sinh x$. (**$2 \sinh^2 x + 1$**)

Solve the equation $\cosh 2x = 3 \sinh x$, giving your answers correct to 2 significant figures. (**0.48, 0.88**)

MB June 1978/1/20 (part)

22. Prove, stating the definition on which your proof is based, that $\cosh^2 x + \sinh^2 x = \cosh 2x$ and that $\cosh^2 x - \sinh^2 x = 1$.

MB November 1977/2/20 (part)

23. Using the formulae $\cosh x = \frac{1}{2}(e^x + e^{-x})$ and $\sinh x = \frac{1}{2}(e^x - e^{-x})$, prove that

- (a) $\cosh^2 x = 1 + \sinh^2 x$
(b) $\cosh 2x = 2\cosh^2 x - 1$

June 1977/1/16 (part)

24. (a) Prove the identity $\cosh^2 u - \sinh^2 u = 1$ and express $\cosh^2 u + \sinh^2 u$ in a simple form.

- (b) Find the only solution of the equation $3 \sinh x - \cosh x = 1$. (**$\ln 2$**)

MB November 1975/2/22 (part)

25. By first expressing $\cosh 2x$ in terms of $\cosh x$, or otherwise, find the real positive root of the equation $\cosh 2x - \cosh x = 2$, giving two significant figures in your answer. (**0.96**)

MB June 1975/1/21

26. Prove that if the equation $a \cosh x + b \sinh x = c$ has distinct real roots for x then $b^2 + c^2 > a^2$.

Solve the equation in the case $a = 2, b = -1, c = 3$. (**1.695, -0.596**)

MB June 1974/2/20

27. (a) Show that $\cosh^2 x - \sinh^2 x = 1$.

- (b) Solve $3 \cosh x - \sinh x = 4$. (**$x = \ln(2 + \sqrt{2})$ or $\ln(2 - \sqrt{2})$**)

MB November 1973/2/22 (part)

28. (a) Show that $\cosh^2 x + \sinh^2 x = \cosh 2x$.

- (b) Solve $4 \cosh x - \sinh x = 8$. (**$x = -\ln 3, \ln 5$**)

MB November 1972/2/21 (part)